Expense Claim Approval System

1. Overview

Build a web application where employees submit expense claims, managers review them, and the finance team gives final approval and records payment. The application is small, but its behaviour is driven entirely by the workflow in this document. Every screen, button and field must follow the rules defined here.
Read the whole document before writing any code. Requirements are spread across sections, and a rule stated in one section applies everywhere.

2. Technical constraints

- Frontend: React or Next.js. TypeScript is preferred.
- Backend: Firebase Authentication (email and password) and Cloud Firestore only. No custom server.
- UI library: your choice.
- Data model: design the Firestore structure yourself. You will be asked to explain your choices.

3. Roles

Role	Description
Employee	Creates, edits and submits their own claims and tracks their status. Every employee has exactly one manager.
Manager	Reviews claims submitted by their direct reports. A manager can also raise their own claims, which are reviewed by Finance instead of a manager.
Finance	Gives second approval on high-value claims and records payment. Finance users do not raise claims.

At signup, the user enters name, email, password and role. A user who signs up as an employee must select their manager from the list of existing managers. Roles cannot be changed after signup.

4. Claim fields

What is CLAIM ? *A financial claim is an employee's or right to receive money from another person, company, or institution.*

Field	Rules
Title	Required, 5 to 80 characters

Field	Rules
Category	Required. One of: Travel, Food, Equipment, Other (must be seeded by default)
Amount (INR)	Required. Greater than 0 and at most 1,00,000
Expense date	Required. Not in the future and not older than 90 days
Description	Required, at least 20 characters (max 5000 characters)
Receipt URL	A valid URL. Required when the amount is 2,000 or more, optional below that

Validation runs before saving a draft as well as before submitting, except that a draft may be saved with only a title.

5. Workflow

A claim is always in exactly one of these states:

State	Meaning
DRAFT	Created by the employee, not yet visible to reviewers
SUBMITTED	Waiting for first review
CHANGES_REQUESTED	Sent back to the owner to edit and resubmit
MANAGER_APPROVED	Passed first review, waiting for Finance
FINANCE_APPROVED	High-value claim approved by Finance, waiting for payment
REJECTED	Final. No further actions
PAID	Final. No further actions

Transitions

From	Action	Performed by	To	Conditions
(new)	Save draft	Owner	DRAFT	Title present
DRAFT	Submit	Owner	SUBMITTED	All field rules pass
DRAFT	Delete	Owner	(removed)	Confirmation dialog required

From	Action	Performed by	To	Conditions
SUBMITTED	Withdraw	Owner	DRAFT	Only while no review action has been taken
SUBMITTED	Approve	First reviewer	MANAGER_APPROVED	Optional comment
SUBMITTED	Request changes	First reviewer	CHANGES_REQUESTED	Comment required
SUBMITTED	Reject	First reviewer	REJECTED	Comment required
CHANGES_REQUESTED	Resubmit	Owner	SUBMITTED	All field rules pass. The claim goes through the full review again
MANAGER_APPROVED	Mark paid	Finance	PAID	Amount below 10,000. Payment reference required
MANAGER_APPROVED	Approve	Finance	FINANCE_APPROVED	Amount 10,000 or more
MANAGER_APPROVED	Reject	Finance	REJECTED	Amount 10,000 or more. Comment required
FINANCE_APPROVED	Mark paid	Finance	PAID	Payment reference required

The first reviewer is the owner's manager for claims raised by employees, and any Finance user for claims raised by managers.

Any action not listed in this table must not be possible from the UI. If a user opens a claim they have no action on, they see it read-only.

6. User flow

6.1 Opening the application

- **Logged out**
  - Opening any page redirects to the Login page.

    - After login, the user returns to the page they originally tried to open, if they are allowed to see it. Otherwise they go to their role's home page (see 6.3).
- **Logged in**
    - Opening the Login or Signup page redirects to the user's home page.
    - The session survives a page refresh. Refreshing must not log the user out or flash the Login page.

6.2 Signup

- **Entry**
    - From the Login page, the user clicks "Create an account" and lands on the Signup page.
- **Form fields**
    - Full name: required.
    - Email: required, must be a valid address.
    - Password: at least 8 characters.
    - Confirm password: must match the password.
    - Role: Employee, Manager or Finance.
    - Errors show under each field once the user leaves it.
- **Manager selection (Employee only)**
    - Selecting Employee shows a required Manager dropdown listing all registered managers by name.
    - Changing the role away from Employee hides the dropdown and clears its value.
    - If no managers are registered yet, the dropdown is replaced by "No managers are registered yet" and Sign up is disabled.

6.3 Login and home pages

- **Login form**
    - Fields: Email and Password, with a Log in button and a link to Signup.
    - Log in is disabled until both fields have a value, and shows a loading state while signing in.
    - Wrong credentials: "Incorrect email or password" appears above the form. Email stays filled; password is cleared.
- **Home page by role**
    - Employee: My Claims
    - Manager: Review Queue, Pending tab
    - Finance: Finance Queue
- **Navigation bar**
    - Shows the user's name, role and a Log out button.
    - Employee links: My Claims
    - Manager links: Review Queue (with pending count badge) and My Claims
    - Finance links: Finance Queue
- **Log out**
    - Returns the user to the Login page.
    - The browser's back button must not show any protected page afterwards.

6.4 Creating and submitting a claim (Employee or Manager)

- **Opening the form**

    - On My Claims, the user clicks New claim and sees an empty claim form.
- **Filling the form**
    - Field errors appear when the user leaves a field, and update as they type after that.
    - Receipt URL is marked optional until the amount is 2,000 or more, then its label changes to required.
- **Save draft**
    - Available once a title is entered. Any other field that has a value must be valid; empty fields are allowed.
    - The claim is created in DRAFT and a history entry is recorded.
    - The user goes to the claim detail page with the message "Draft saved". The claim appears in My Claims with a Draft badge.
- **Submit**
    - Available on a new form (the claim goes straight to SUBMITTED) and on an existing draft.
    - The whole form is validated. If any rule fails, the form scrolls to the first error and nothing is saved.
    - If valid, a confirmation dialog shows the title, amount and reviewer: the employee's manager by name, or "Finance team" for a manager's own claim.
    - On confirm: the claim moves to SUBMITTED, a history entry is recorded, and the user goes to the claim detail page with "Claim submitted".
    - The claim appears in the reviewer's queue immediately.
- **Managing a draft**
    - Edit reopens the form with the saved values.
    - Delete opens a confirmation dialog. On confirmation, the claim is removed and the user returns to My Claims.

6.5 Tracking a claim (owner)

- **My Claims list**
    - Shows the user's claims with the fields listed in section 8 and a status badge.
    - Updates live as reviewers act.
- **Claim detail page**
    - Shows every claim field, the current status, and a timeline of all actions taken on the claim.
    - While the claim is still in progress, shows a "Currently with" line that tells the owner who needs to act next:
        - SUBMITTED: the reviewing manager's name. If the claim was submitted by a manager, it shows "Finance team" instead.
        - CHANGES_REQUESTED: "You" (the owner must update and resubmit the claim)
        - MANAGER_APPROVED or FINANCE_APPROVED: "Finance team"
- **Withdrawing a submitted claim**
    - While SUBMITTED, the owner sees the claim read-only with a Withdraw button.
    - Withdraw opens a confirmation dialog. On confirmation, the claim returns to DRAFT, leaves the reviewer's queue and becomes editable again.
    - If the reviewer acts first, the withdrawal fails with "This claim has already been reviewed" and the page shows the new state.
- **Final states**
    - REJECTED: the rejection reason is shown prominently at the top of the detail page.
    - PAID: the payment reference and paid date are shown.
    - Neither has any action.

6.6 Reviewing claims (Manager)

- **Pending tab**
  - The manager lands here after login. The navigation badge shows the pending count and updates live.
  - Lists every SUBMITTED claim from the manager's direct reports, oldest submission first.
  - Each row shows employee name, title, category, amount, expense date, submitted date and days waiting.
  - Filters: employee and category.
  - Empty state: "Nothing to review".
  - Never shows drafts, the manager's own claims, or claims of employees who report to another manager.
- **Opening a claim**
  - Clicking a row opens the claim detail page.
  - Shows all fields, the receipt link (opens in a new tab), the employee's name and the full history, including earlier change requests and resubmissions.
  - Shows three actions: Approve, Request changes and Reject.
- **Actions**
  - Approve:
    - Dialog with an optional comment.
    - On confirmation, the claim moves to MANAGER_APPROVED, a history entry is recorded with the comment, and the claim appears in the Finance Queue.
  - Request changes:
    - Dialog with a required comment. Confirm stays disabled until the comment has text other than spaces.
    - On confirmation, the claim moves to CHANGES_REQUESTED and returns to the owner.
  - Reject:
    - Dialog with a required comment and the warning "Rejection is final. The employee cannot resubmit this claim."
    - On confirmation, the claim moves to REJECTED.
- **After an action**
  - The manager sees a message for that action and returns to the Pending tab.
  - The claim leaves Pending and appears in the Reviewed tab.
- **Reviewed tab**
  - Lists claims this manager has acted on, most recent action first.
  - Shows the action taken, action date and the claim's current status, so the manager can see when Finance later pays or rejects it.
- **Resubmissions and conflicts**
  - A resubmitted claim reappears in Pending and goes through the full review again.
  - If the claim is withdrawn or changes state while open, the action buttons disappear immediately and the page shows the new state.
  - If the manager confirms an action at that same moment, it fails with "This claim has already been updated" (Rule 6).

6.7 Responding to a change request (owner)

- **Noticing it**
  - The claim shows a Changes requested badge in My Claims and counts towards

"needs your action" in the summary.
    - The detail page shows the reviewer's latest comment at the top, highlighted, with an Edit button.
- **Editing**
    - Edit opens the form with current values filled in and the reviewer's comment above it.
    - Save changes saves the edits; the claim stays in CHANGES_REQUESTED.
- **Resubmitting**
    - Resubmit validates the whole form and shows the same confirmation dialog as Submit.
    - On confirmation, the claim moves to SUBMITTED, a history entry is recorded and the claim reappears in the reviewer's Pending tab.

6.8 Finance processing

- **Finance Queue**
    - The finance user lands here after login. Each tab shows its count in the tab label.
    - Any Finance user can act on any claim here, including one where the same user did the first review.
- **First review tab**
    - Lists SUBMITTED claims raised by managers, oldest first.
    - Same detail page and same three actions as a manager review (see 6.6, Actions).
    - Approving moves the claim to MANAGER_APPROVED; the history records the Finance user as the actor.
- **Needs approval tab**
    - Lists MANAGER_APPROVED claims of 10,000 or more, oldest approval first.
    - Approve (optional comment): moves the claim to FINANCE_APPROVED and into Ready for payment.
    - Reject (required comment): moves the claim to REJECTED.
- **Ready for payment tab**
    - Lists MANAGER_APPROVED claims below 10,000 and all FINANCE_APPROVED claims.
    - Shows the total amount of the listed claims above the list.
    - Mark paid opens a dialog asking for a payment reference (required, for example a bank transaction ID).
    - On confirmation, the claim moves to PAID, the payment reference and paid date are saved, and a history entry is recorded.
- **Paid tab**
    - Lists PAID claims with owner, amount, payment reference and paid date.
    - Filter by month, with the total paid for the selected month.

6.9 Who can see a claim

User	Can open
Owner	All of their own claims, in any state
Manager	Claims of their direct reports in any state except DRAFT, plus their own claims

User	Can open
Finance	Claims raised by managers in any state except DRAFT, and any claim that has reached MANAGER_APPROVED or later

Opening any other claim, or a page meant for another role (for example an employee opening the Review Queue), shows the "not authorised" page with a link back to the user's home page.

6.10 Errors and edge cases

- **Duplicate actions**
  - Every action button is disabled while its request is in progress, so a double click cannot submit or approve twice.
- **Network failure**
  - The user sees an error message.
  - The claim keeps its previous state and the form keeps the entered values.
  - The button becomes clickable again.
- **Invalid URLs**
  - A deleted claim's URL shows "This claim no longer exists" with a link back home.
  - REJECTED and PAID claims are read-only for everyone, including through the edit page URL.

6.11 Example: one claim from start to finish

Riya is an employee who reports to Arjun (manager). Meera is a Finance user.

Step	Who	Action	Result
1	Riya	Creates a Travel claim for 12,500 with a receipt URL and saves a draft	DRAFT. Visible only to Riya
2	Riya	Submits the draft	SUBMITTED. Appears in Arjun's Pending tab
3	Arjun	Requests changes: "Attach the hotel invoice, not the booking page"	CHANGES_REQUESTED. Riya sees the comment
4	Riya	Edits the receipt URL and resubmits	SUBMITTED. Back in Arjun's Pending tab
5	Arjun	Approves	MANAGER_APPROVED. Appears in Meera's Needs approval tab because the amount is 10,000 or more
6	Meera	Approves	FINANCE_APPROVED. Moves to Ready for payment
7	Meera	Marks paid with reference UTR4471920	PAID. Riya sees the reference. The timeline shows all six actions

Had the amount been 8,000, step 6 would be skipped: after Arjun's approval the claim would go straight to Ready for payment.

8. Screens

- Refer to the link below for the wireframes. They show the flow of each screen only and are not design mockups. The visual and UI design is left to your own judgement.

  https://claude.ai/artifact/3FBkQh7AmRZbwCNdVeuYHP#page-dcb81aee8d83